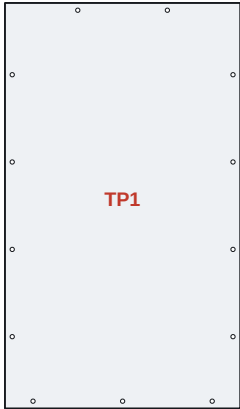
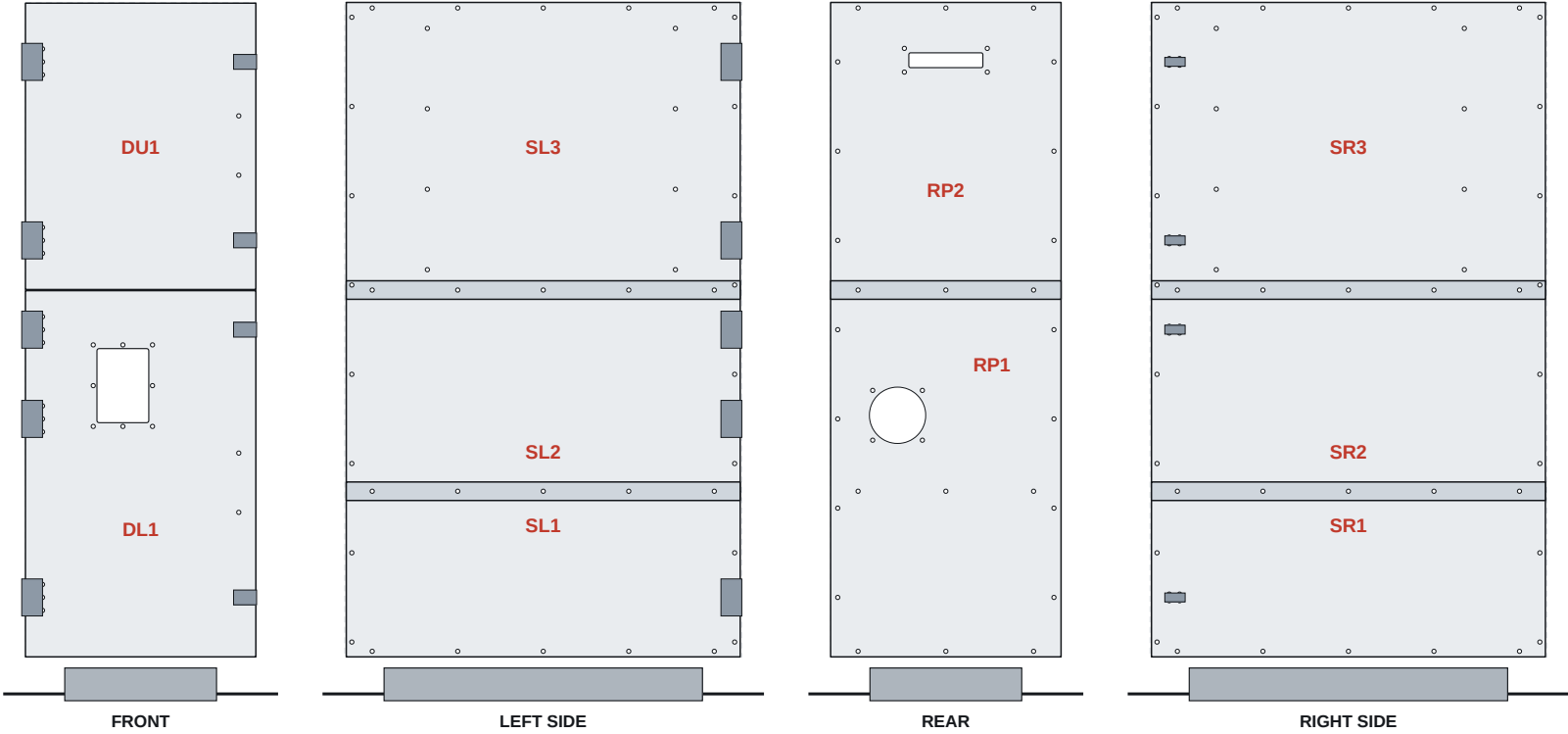


SKIN LAYOUT - EXTERIOR ELEVATIONS 1:20 (pieces as seen from outside)



Grey bars: hinges (front + left), latches + keepers (front + right).  
Strips JS/JR cover the skin joints on the mid and shelf rails.  
Skirts SK1-SK3 sit under the base rails between the castor pads.  
Dots: D8 screw holes on the frame rivnuts (weld pack sheet 6).  
Every frame rivnut is matched by a hole (checked by sheetmetal.py).

NOTES

- 1 DXF (cad/dxf/cut) is the master geometry, 1:1 mm. info/ adds bend lines, ID etch and notes on layers.
- 2 Skins 1.2 mm steel (Zincanneal/CR4), powder coat both sides, then bond 3 mm MLV inside (D8 at the frame screws: it is the gasket).
- 3 Frame screws: M6 x 20 flanged button head through D8 (+1 mm frame tolerance) into the frame rivnuts. Rivets: 4.8 mm blind, D5 holes. Deburr all holes; mask nothing - coat before rivnuts.
- 4 Folded parts: flat blanks for R = t inside, K 0.33. Check one test bend on your brake first.
- 5 Split skins meet on the rail centreline with a 1 mm gap and R4 notches round each strip screw; the strip clamps both edges (sealant under). Detail J, sheet 5.

PARTS (one DXF each)

| ID  | QTY | PART                                 | t   | FLAT       | HOLES | BENDS |
|-----|-----|--------------------------------------|-----|------------|-------|-------|
| SL1 | 1   | Side skin left - bottom - HOLD M3    | 1.2 | 1064 x 446 | 12    | 0     |
| SL2 | 1   | Side skin left - middle - HOLD M3    | 1.2 | 1064 x 542 | 10    | 0     |
| SL3 | 1   | Side skin left - top                 | 1.2 | 1064 x 776 | 27    | 0     |
| SR1 | 1   | Side skin right - bottom - HOLD M3   | 1.2 | 1064 x 446 | 13    | 0     |
| SR2 | 1   | Side skin right - middle - HOLD M3   | 1.2 | 1064 x 542 | 8     | 0     |
| SR3 | 1   | Side skin right - top                | 1.2 | 1064 x 776 | 29    | 0     |
| JS1 | 2   | Joint strip, side skins - mid rail   | 1.2 | 1064 x 50  | 5     | 0     |
| JS2 | 2   | Joint strip, side skins - shelf rail | 1.2 | 1064 x 50  | 7     | 0     |
| RP1 | 1   | Rear skin - bottom - HOLD M3         | 1.2 | 622 x 990  | 19    | 0     |
| RP2 | 1   | Rear skin - top                      | 1.2 | 622 x 776  | 13    | 0     |
| JR1 | 1   | Joint strip, rear skin               | 1.2 | 622 x 50   | 3     | 0     |
| TP1 | 1   | Top skin                             | 1.2 | 622 x 1072 | 13    | 0     |
| DL1 | 1   | Door, lower                          | 1.2 | 622 x 988  | 23    | 0     |
| DU1 | 1   | Door, upper                          | 1.2 | 622 x 773  | 12    | 0     |
| DA1 | 1   | Stiffener lower door, left vertical  | 1.2 | 941 x 38   | 9     | 1     |
| DA2 | 1   | Stiffener lower door, right vertical | 1.2 | 941 x 38   | 6     | 1     |
| DA3 | 1   | Stiffener upper door, left vertical  | 1.2 | 727 x 38   | 6     | 1     |
| DA4 | 1   | Stiffener upper door, right vertical | 1.2 | 727 x 38   | 6     | 1     |
| DA5 | 4   | Stiffener horizontal                 | 1.2 | 510 x 38   | 0     | 1     |
| KB1 | 4   | Latch keeper bracket                 | 1.2 | 83 x 40    | 2     | 1     |
| WR1 | 1   | Window retainer frame                | 1.2 | 180 x 240  | 8     | 0     |
| SK1 | 1   | Plinth skirt, front                  | 1.2 | 410 x 116  | 3     | 1     |
| SK2 | 1   | Plinth skirt, rear                   | 1.2 | 410 x 116  | 3     | 1     |
| SK3 | 2   | Plinth skirt, side                   | 1.2 | 860 x 116  | 4     | 1     |
| DT1 | 1   | Drip tray                            | 1.2 | 603 x 1053 | 1     | 4     |
| PD1 | 4   | Castor pad                           | 8   | 100 x 100  | 4     | 0     |
| PC1 | 4   | Post top cap                         | 3   | 30 x 30    | 0     | 0     |
| RR1 | 4   | 19in rail, 16U                       | 2   | 721 x 46   | 51    | 1     |

46 pieces, 78 kg. Largest blank 1064 x 776 (limit 1200 x 800).

Blanks 1200x800: steel\_1.2mm x11, SS304\_1.2mm x1, plate\_8mm x1, plate\_3mm x1, steel\_2mm x1

ORDERING

Send cad/dxf/cut/ - each file is named ID\_thickness\_material\_qty,  
e.g. SL3\_1.2mm\_steel\_x1.dxf - plus bom/sheet\_metal\_parts.csv.  
Or cut the nested blanks in cad/dxf/nest/ on a 1200x800 bed.  
Folding: 11 parts, 24 bends (flat blanks: R = t, K 0.33).  
Tap the four PD1 pads M8 after cutting. Coat after folding.  
HOLD RP1 SL1 SL2 SR1 SR2 until tape check M3: they move with the mid rail.

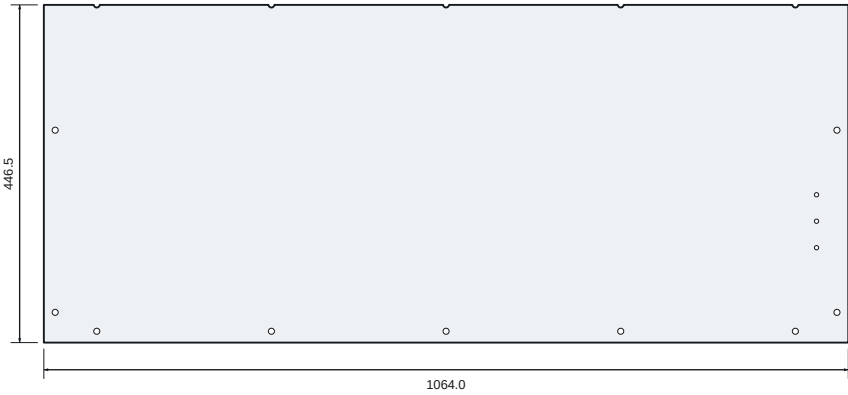
|         |                                                             |
|---------|-------------------------------------------------------------|
| PROJECT | SRA-16 Silent AC Rack - laser-cut sheet-metal parts         |
| TITLE   | Skin layout, parts list                                     |
| DOC NO. | CP-SRA16-SMP-001 SHEET 1 OF 6 REV A                         |
| SCALE   | 1:20 (elevations) @ A3 UNITS mm LAYOUT v1.2.0               |
| DATE    | 2026-09-27 DRAWN Claude (AI) for Carbon Project             |
| STATUS  | FOR CUTTING - DXF is the master; confirm hardware (sheet 5) |

SL1 x1

Side skin left - bottom

1:10

Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



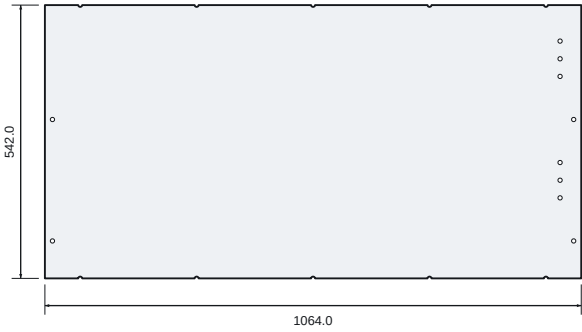
- 9 x D8 M6 screw to frame rivnut
- 3 x D5 rivet, hinge frame leaf
- 5 x R4 edge notch, joint strip screw

SL2 x1

Side skin left - middle

1:15

Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



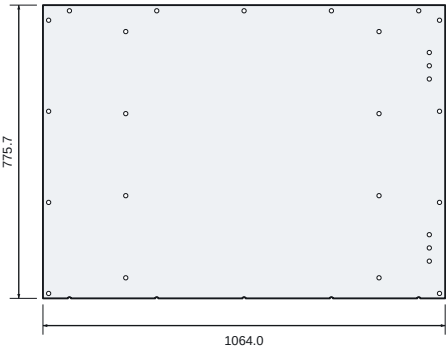
- 4 x D8 M6 screw to frame rivnut
- 6 x D5 rivet, hinge frame leaf
- 10 x R4 edge notch, joint strip screw

SL3 x1

Side skin left - top

1:20

Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



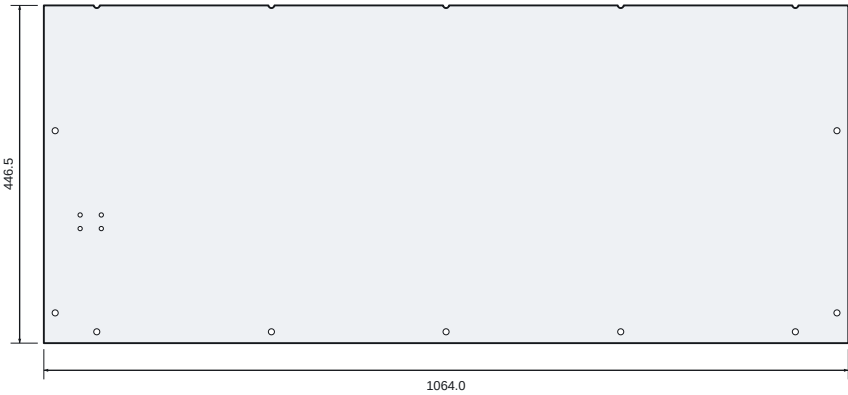
- 21 x D8 M6 screw to frame rivnut
- 6 x D5 rivet, hinge frame leaf
- 5 x R4 edge notch, joint strip screw

SR1 x1

Side skin right - bottom

1:10

Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



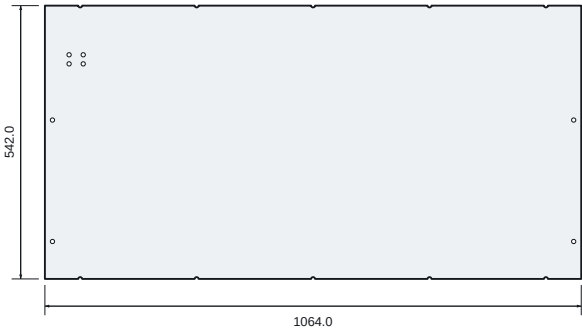
- 9 x D8 M6 screw to frame rivnut
- 4 x D5 rivet, toggle latch base
- 5 x R4 edge notch, joint strip screw

SR2 x1

Side skin right - middle

1:15

Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



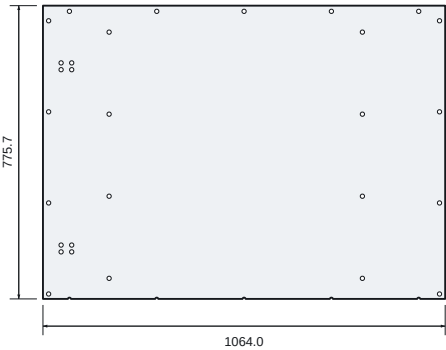
- 4 x D8 M6 screw to frame rivnut
- 4 x D5 rivet, toggle latch base
- 10 x R4 edge notch, joint strip screw

SR3 x1

Side skin right - top

1:20

Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



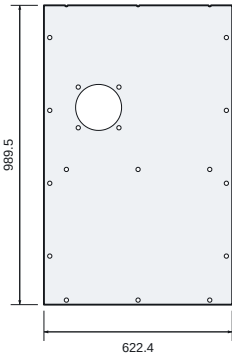
- 21 x D8 M6 screw to frame rivnut
- 8 x D5 rivet, toggle latch base
- 5 x R4 edge notch, joint strip screw

RP1 x1

Rear skin - bottom

1:25

Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



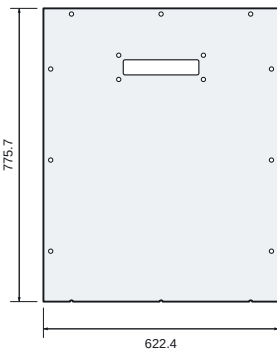
- 14 x D8 M6 screw to frame rivnut
- 1 x D152 exhaust duct D150
- 4 x D5.5 M5, exhaust spigot flange
- 3 x R4 edge notch, joint strip screw

RP2 x1

Rear skin - top

1:20

Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



- 9 x D8 M6 screw to frame rivnut
- 4 x D4.5 M4, brush cable grommet
- cable entry 200x40
- 3 x R4 edge notch, joint strip screw

TP1 x1

Top skin

1:25

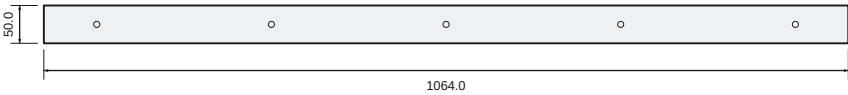
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



- 13 x D8 M6 screw to frame rivnut

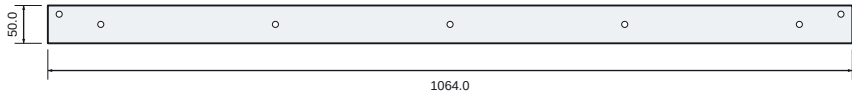
|         |                                                             |
|---------|-------------------------------------------------------------|
| PROJECT | SRA-16 Silent AC Rack - laser-cut sheet-metal parts         |
| TITLE   | Flat patterns - side, rear and top skins                    |
| DOC NO. | CP-SRA16-SMP-001 SHEET 2 OF 6 REV A                         |
| SCALE   | as noted per part @ A3 UNITS mm LAYOUT v1.2.0               |
| DATE    | 2026-09-27 DRAWN Claude (AI) for Carbon Project             |
| STATUS  | FOR CUTTING - DXF is the master; confirm hardware (sheet 5) |

**JS1 x2** Joint strip, side skins - mid rail 1:10  
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



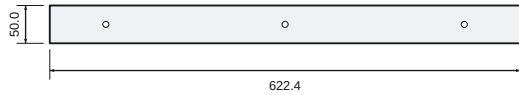
5 x D8 M6 screw to frame rivnut

**JS2 x2** Joint strip, side skins - shelf rail 1:10  
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



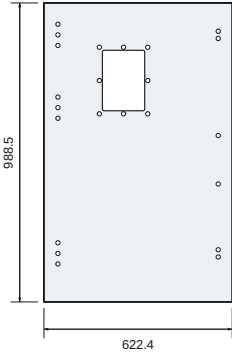
7 x D8 M6 screw to frame rivnut

**JR1 x1** Joint strip, rear skin 1:10  
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



3 x D8 M6 screw to frame rivnut

**DL1 x1** Door, lower (AC bay) 1:25  
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



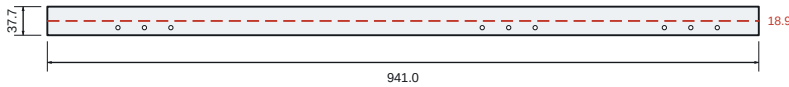
9 x D5 rivet, hinge door leaf  
4 x D5 rivet, latch keeper KB1  
2 x D5.5 M5, pull handle  
8 x D4.5 M4, window clamp

**DU1 x1** Door, upper (rack) 1:20  
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



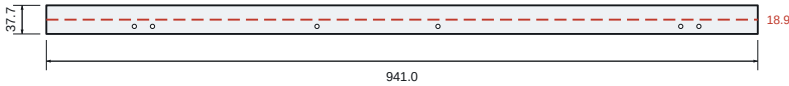
6 x D5 rivet, hinge door leaf  
4 x D5 rivet, latch keeper KB1  
2 x D5.5 M5, pull handle

**DA1 x1** Door stiffener, lower door, left vertical 1:10  
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



9 x D5 rivet, hinge

**DA2 x1** Door stiffener, lower door, right vertical 1:10  
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



4 x D5 rivet, latch keeper  
2 x D5.5 M5, pull handle screw

**DA3 x1** Door stiffener, upper door, left vertical 1:10  
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



6 x D5 rivet, hinge

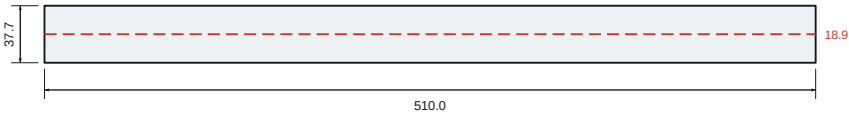
**DA4 x1** Door stiffener, upper door, right vertical 1:10  
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



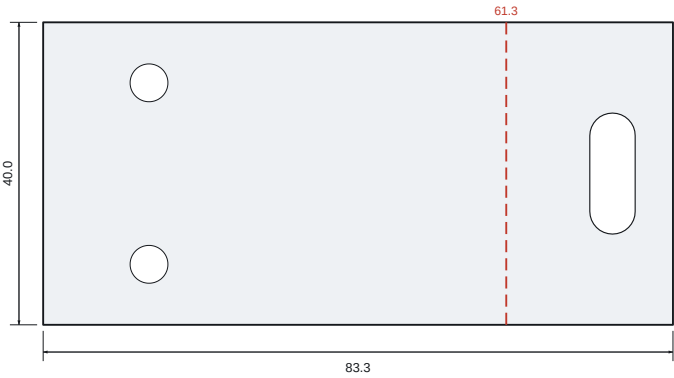
4 x D5 rivet, latch keeper  
2 x D5.5 M5, pull handle screw

|         |                                                             |
|---------|-------------------------------------------------------------|
| PROJECT | SRA-16 Silent AC Rack - laser-cut sheet-metal parts         |
| TITLE   | Flat patterns - doors, joint strips, door stiffeners        |
| DOC NO. | CP-SRA16-SMP-001 SHEET 3 OF 6 REV A                         |
| SCALE   | as noted per part @ A3 UNITS mm LAYOUT v1.2.0               |
| DATE    | 2026-09-27 DRAWN Claude (AI) for Carbon Project             |
| STATUS  | FOR CUTTING - DXF is the master; confirm hardware (sheet 5) |

**DA5 x4** Door stiffener, horizontal (top/bottom, both doors) 1:5  
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035

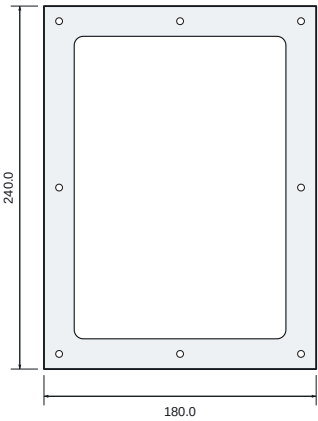


**KB1 x4** Latch keeper bracket 1:1  
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



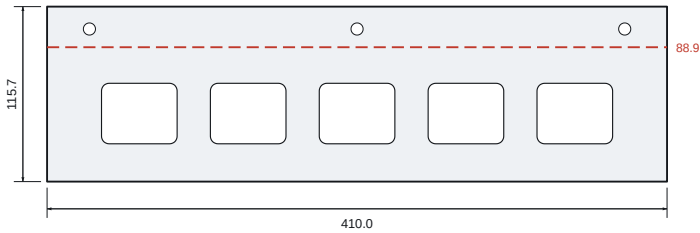
2 x D5 rivet to door + stiffener  
latch catch slot

**WR1 x1** Window retainer frame 1:5  
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



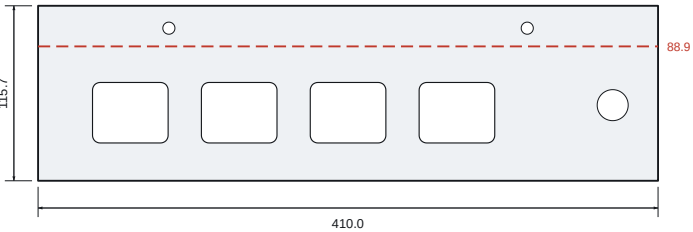
8 x D4.5 M4, window clamp  
window opening

**SK1 x1** Plinth skirt, front 1:5  
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



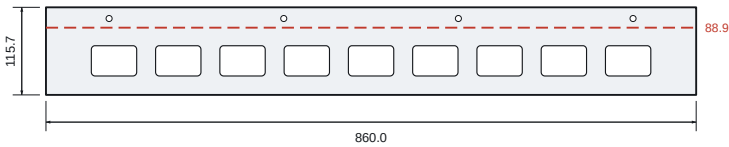
3 x D8 M6 screw up into base-rail rivnut  
air slot (x5)

**SK2 x1** Plinth skirt, rear (drain outlet) 1:5  
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



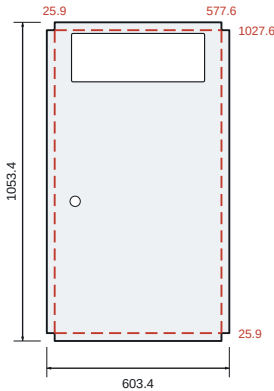
2 x D8 M6 screw up into base-rail rivnut  
1 x D20.5 drain barb bulkhead  
air slot (x4)

**SK3 x2** Plinth skirt, side (right: same part turned round) 1:10  
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



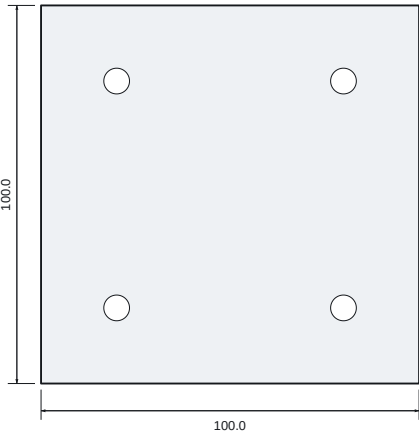
4 x D8 M6 screw up into base-rail rivnut  
air slot (x9)

**DT1 x1** Drip tray (304 stainless pan) 1:25  
Stainless 304 sheet, 2B 1.2 - as cut, TIG-weld corners, passivate



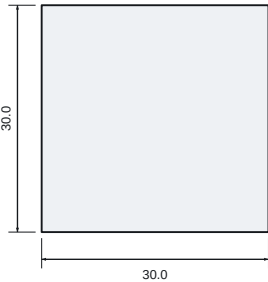
1 x D34 25 mm tank bulkhead  
room-air inlet (riser)

**PD1 x4** Castor pad (weld pack mark PL1) 1:2  
Steel plate (grade 250) 8.0 - welded to frame, coated with it

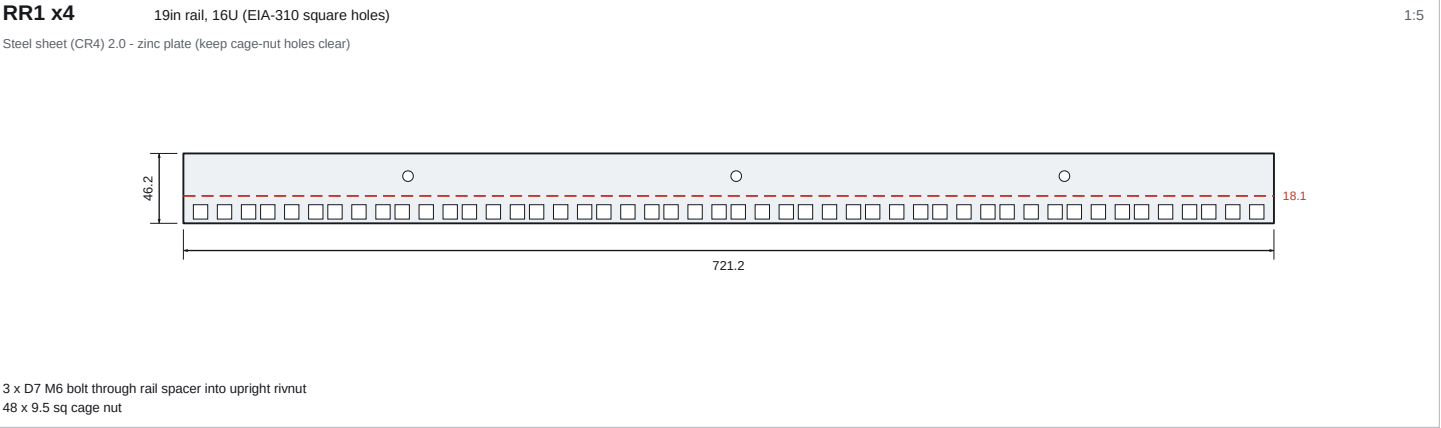


4 x D6.8 tap M8 through

**PC1 x4** Post top cap (weld pack mark PL2) 1:1  
Steel plate (grade 250) 3.0 - welded to frame, coated with it



|         |                                                                   |
|---------|-------------------------------------------------------------------|
| PROJECT | SRA-16 Silent AC Rack - laser-cut sheet-metal parts               |
| TITLE   | Flat patterns - stiffeners, keepers, window, skirts, tray, plates |
| DOC NO. | CP-SRA16-SMP-001 SHEET 4 OF 6 REV A                               |
| SCALE   | as noted per part @ A3 UNITS mm LAYOUT v1.2.0                     |
| DATE    | 2026-09-27 DRAWN Claude (AI) for Carbon Project                   |
| STATUS  | FOR CUTTING - DXF is the master; confirm hardware (sheet 5)       |



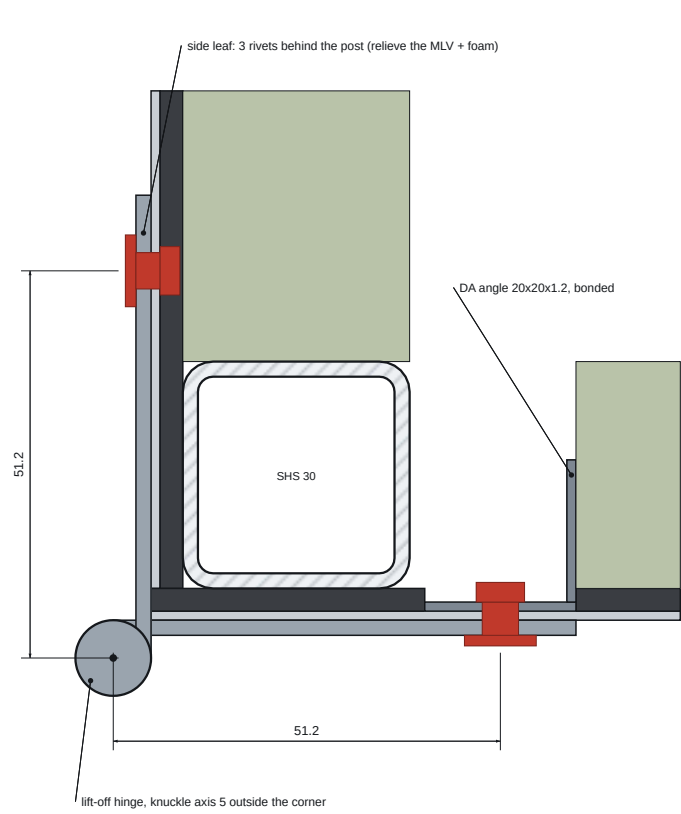
HARDWARE - hole patterns assume these parts (confirm before cutting)

| ITEM                                  | QTY | HOLES / NOTE                                                                                                              |
|---------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------|
| Lift-off butt hinge, stainless        | 5   | 100 long; rivet line 51.2 mm from the knuckle axis on both leaves (drill the leaves to suit), 3 x D5 per leaf at 35 pitch |
| Adjustable toggle latch, stainless    | 4   | base 4 x D5 on 28 x 18, first row 52.2 mm behind the door face (detail L); hook engages the KB1 slot                      |
| Latch keeper KB1 (made, sheet 4)      | 4   | riveted to door + stiffener, 2 x D5                                                                                       |
| Pull handle 160 c/c                   | 2   | 2 x M5 from inside, through door + DA stiffener                                                                           |
| Blind rivet 4.8 mm, stainless         | 54  | 4.8 x 8 (grip 3-5) for hinges and KB1, 4.8 x 6 for the latch bases; grips: door leaf 4.4, side leaf 3.2, KB1 3.6          |
| M6 x 20 flanged button head           | 140 | skins, strips and skirts into the frame rivnuts                                                                           |
| Polycarbonate 6 mm, 180x240           | 2   | drilled 8 x D5 to the WR1 pattern                                                                                         |
| EPDM 10 mm spacer frame + 8 x M4 x 35 | 1   | window clamp: skin / pane / spacer / pane / WR1                                                                           |
| Exhaust flanged spigot D150           | 1   | 4 x M5 on PCD 190 at 45 deg (RP1)                                                                                         |
| Brush cable grommet 200 x 40          | 1   | 4 x M4 on 224 x 64 (RP2)                                                                                                  |
| 16 mm drain bulkhead barb             | 1   | D20.5 in SK2                                                                                                              |

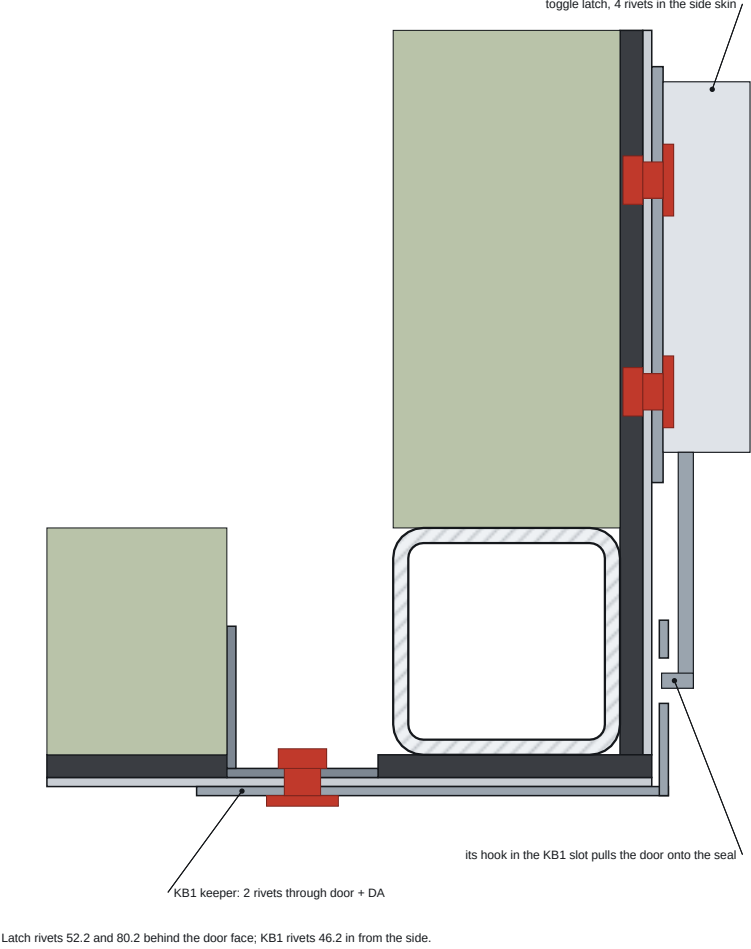
ASSEMBLY

- 1 Coat all parts. Bond MLV inside the skins and doors: punch it D8 at the frame screws (it is the gasket), clear it 3 mm round rivets and M4/M5 holes.
- 2 Doors: bond the DA stiffener frame inside (legs inboard, 2 mm inside the frame opening), rivet the hinges and KB1 keepers through skin + angle, fit handles.
- 3 Lower door: glazing unit - outer pane, EPDM spacer, inner pane, WR1 - clamped with 8 x M4 from outside.
- 4 Side skins bottom to top, then the JS strips over the joints (acoustic sealant under each strip edge).
- 5 Rear skins + JR strip, exhaust spigot and cable grommet; top skin last (it overlaps the side and rear skins).
- 6 Skirts SK1-SK3 up into the base-rail rivnuts. Hang the doors on the lift-off hinges, rivet the latches to the right-hand skins and adjust them to pull the doors 3 mm onto the seal.

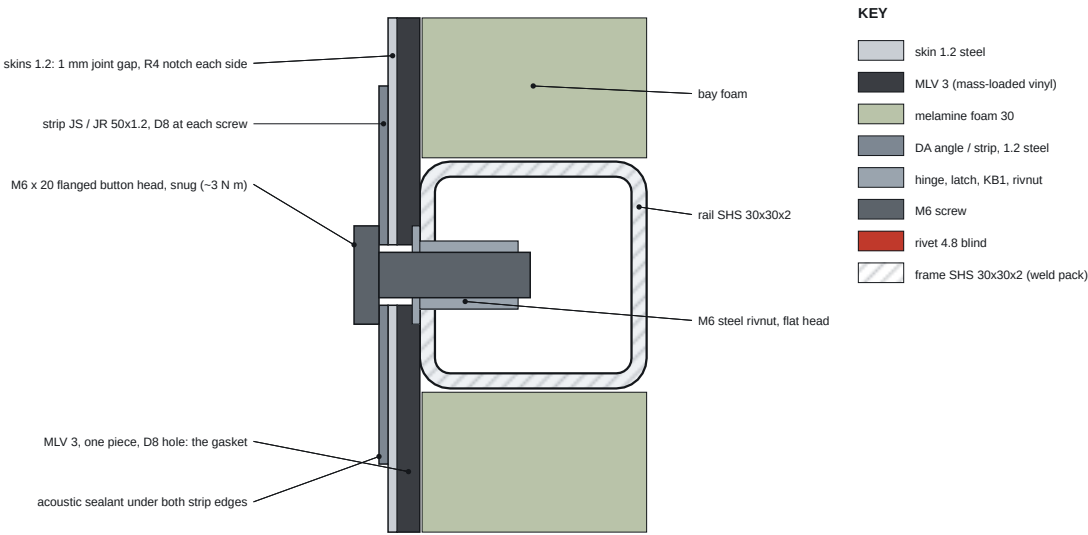
H HINGE CORNER - PLAN 1:1



L LATCH CORNER - PLAN 1:1

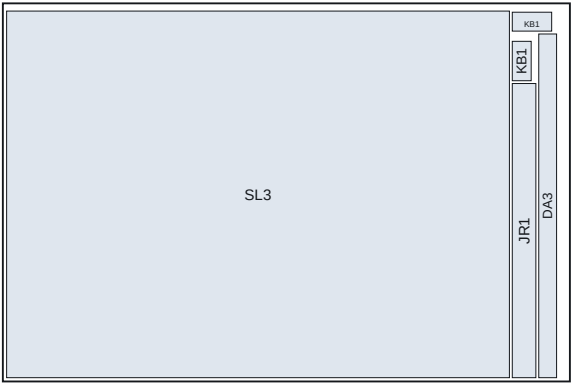


J SKIN JOINT ON A RAIL - SECTION THROUGH A STRIP SCREW 1:1

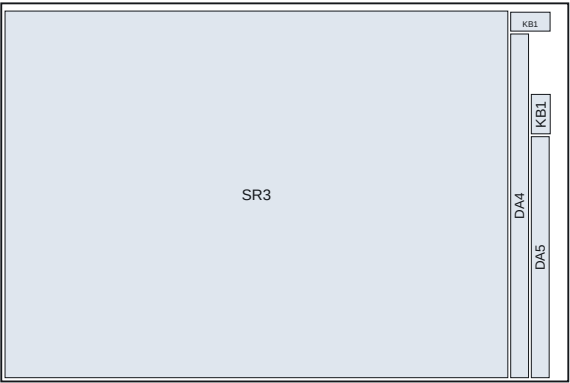


|         |                                                             |
|---------|-------------------------------------------------------------|
| PROJECT | SRA-16 Silent AC Rack - laser-cut sheet-metal parts         |
| TITLE   | 19in rail, hardware, assembly, details                      |
| DOC NO. | CP-SRA16-SMP-001 SHEET 5 OF 6 REV A                         |
| SCALE   | as noted @ A3 UNITS mm LAYOUT v1.2.0                        |
| DATE    | 2026-09-27 DRAWN Claude (AI) for Carbon Project             |
| STATUS  | FOR CUTTING - DXF is the master; confirm hardware (sheet 5) |

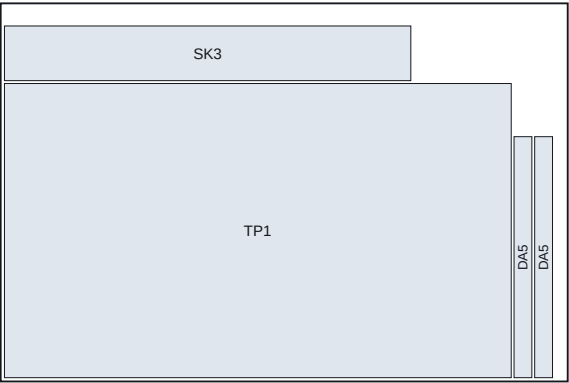
NESTING ON 1200 x 800 BLANKS (MaxRects, 6 mm gap, 8 mm margin) - cad/dxf/nest/



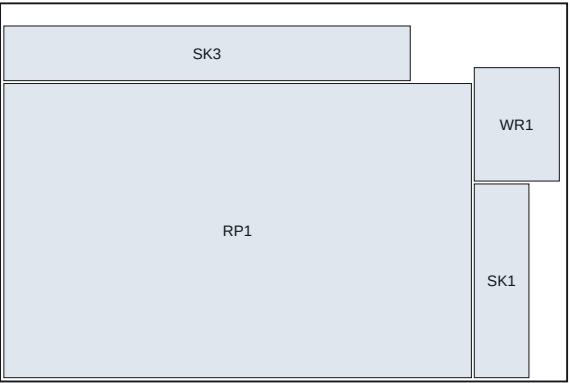
steel 1.2mm sheet 1 - 93% used



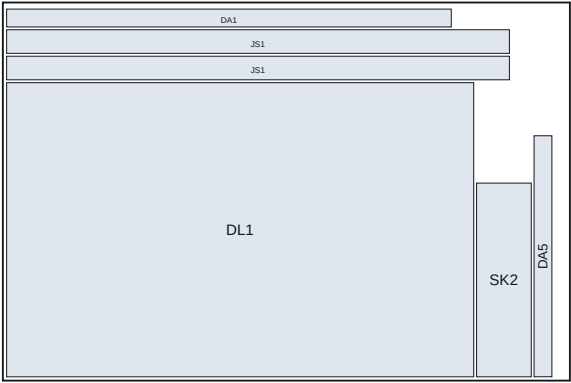
steel 1.2mm sheet 2 - 92% used



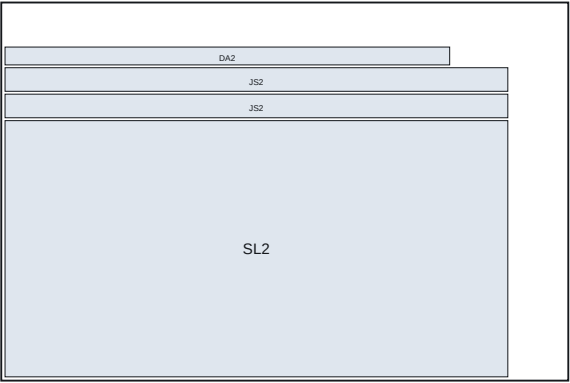
steel 1.2mm sheet 3 - 84% used



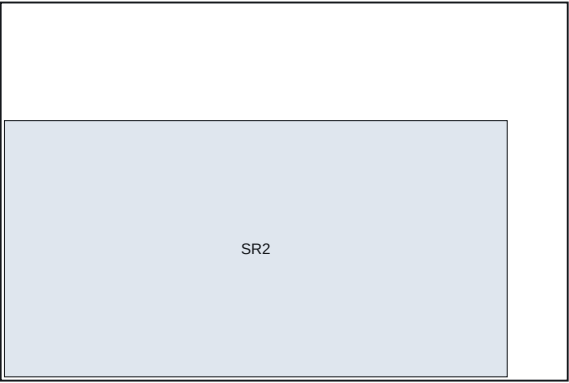
steel 1.2mm sheet 4 - 84% used



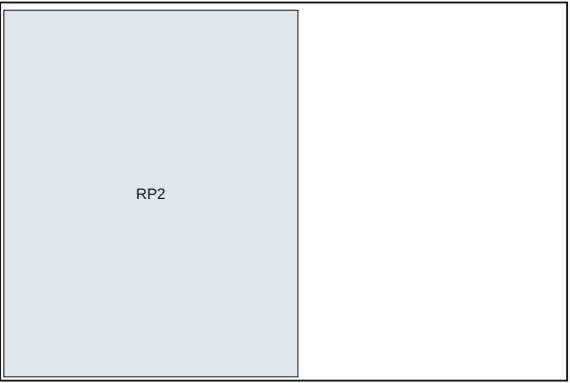
steel 1.2mm sheet 5 - 86% used



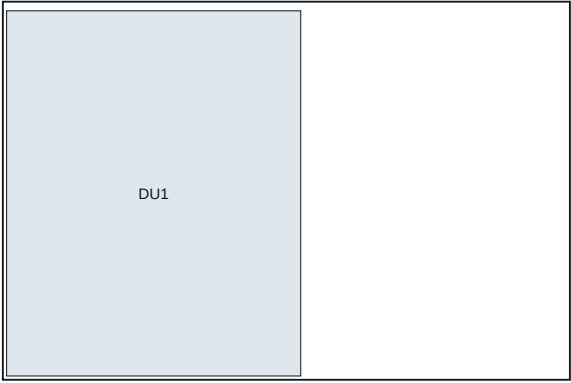
steel 1.2mm sheet 6 - 75% used



steel 1.2mm sheet 7 - 60% used



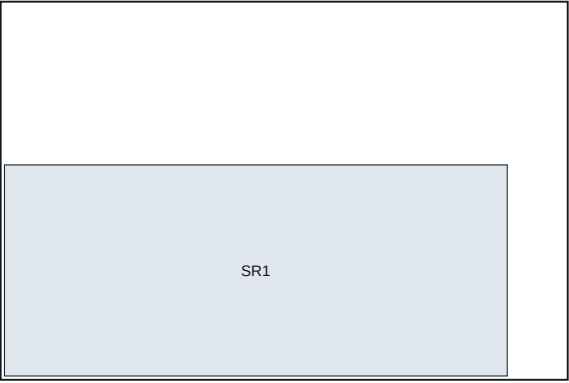
steel 1.2mm sheet 8 - 50% used



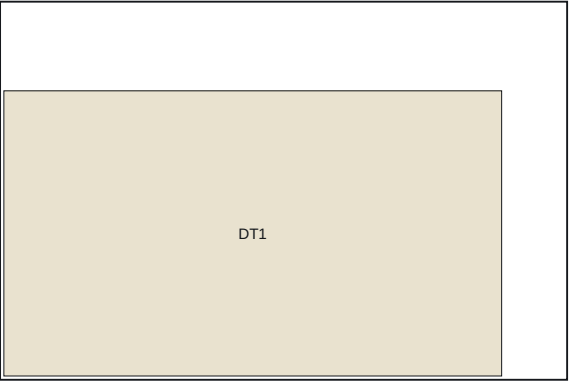
steel 1.2mm sheet 9 - 50% used



steel 1.2mm sheet 10 - 49% used



steel 1.2mm sheet 11 - 49% used



SS304 1.2mm sheet 1 - 66% used

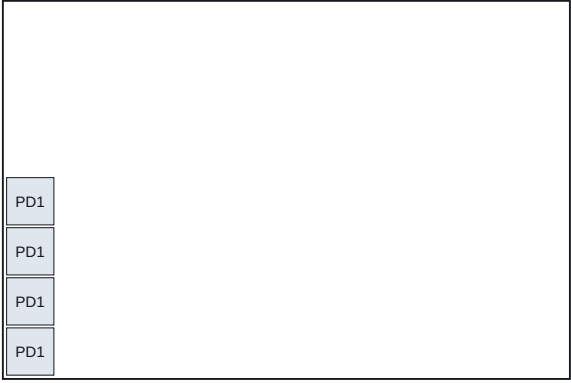


plate 8mm sheet 1 - 4% used

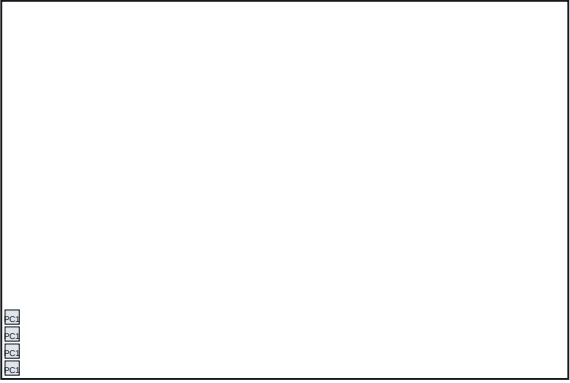
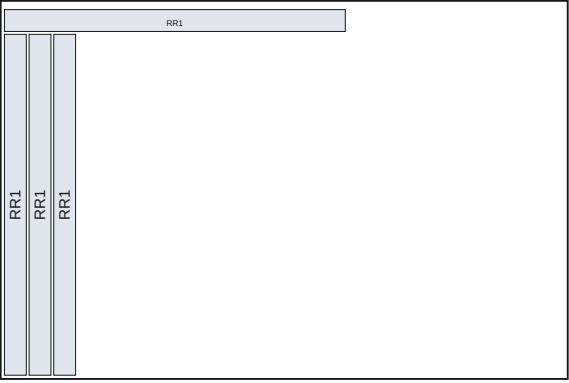


plate 3mm sheet 1 - 0% used



steel 2mm sheet 1 - 14% used

NOTES

Blanks are the 1200 x 800 sheet / bed size. Each nest DXF has the parts on CUT and the blank on SHEET.

11 steel 1.2 blanks: the big skins cannot share a blank (no two fit together), so their offcuts carry the strips, stiffeners, keepers and skirts.

PD1 (8 mm), PC1 (3 mm) and RR1 (2 mm) are small: cut them from offcuts or order them as cut parts; their nests are shown for completeness.

Parts may turn 90 deg (no grain on coated steel). 6 mm between parts, 8 mm from the blank edge.

|         |                                                             |
|---------|-------------------------------------------------------------|
| PROJECT | SRA-16 Silent AC Rack - laser-cut sheet-metal parts         |
| TITLE   | Nesting on 1200 x 800 blanks                                |
| DOC NO. | CP-SRA16-SMP-001 SHEET 6 OF 6 REV A                         |
| SCALE   | 1:16 @ A3 UNITS mm LAYOUT v1.2.0                            |
| DATE    | 2026-09-27 DRAWN Claude (AI) for Carbon Project             |
| STATUS  | FOR CUTTING - DXF is the master; confirm hardware (sheet 5) |